Tianren (Tyler) Zeng

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Summary

UC Berkeley EECS junior focused on robotics, computer vision, and machine learning. Interested in autonomous systems, ROS2-based robot software, UAV perception and control, humanoid robot learning, and practical AI tools that turn domain workflows into usable software.

Education

University of California, Berkeley | Berkeley, CA | EECS, Robotics Focus | Expected May 2027
Junior standing in Electrical Engineering and Computer Sciences with a technical focus on robotics, computer vision, and machine learning.

Irvine Valley College | Irvine, CA | Computer Science | May 2025
Completed lower-division coursework before transferring to UC Berkeley.

Relevant Coursework

Spring 2026: CS 189 Machine Learning | CS 188 Introduction to Artificial Intelligence | EECS 106A Introduction to Robotics | CS 61B Data Structures
Fall 2026: EECS 127 Optimization Models in Engineering | ELENG C128 Feedback Control Systems

Skills

Programming: Python, Java, C++, C
Robotics: ROS2 coding (Python), humanoid robot simulation environment setup, UAV drone simulation environment setup
AI / Vision: Machine learning, computer vision, OpenCV, reinforcement learning
Web / Software: TypeScript, React, Next.js, Tailwind CSS

Research / Publication

Agents' Last Exam | arXiv:2606.05405 | Co-author | 2026 | arXiv
- Co-authored a benchmark paper evaluating AI agents on long-horizon, economically valuable professional workflows with verifiable outcomes.

Project Experience

Agent-HLE Task Proposal Agent | Next.js, TypeScript, React, Fuse.js, OpenRouter | 2026 | GitHub
- Built a web app that maps user professional backgrounds to Agent-HLE benchmark landscapes using keyword matching with LLM fallback, then generates customized task proposals.
- Implemented streaming proposal generation, review and refine workflows, and API-side safeguards such as rate limiting and structured fallback outputs.
- Outcome: delivered an end-to-end tool that turns a single user background prompt into benchmark-aligned task proposals with structured outputs and review support.

BYOW Procedural Dungeon Game (CS61B Project 5) | Java | Spring 2026 | GitHub
- Developed a tile-based world generator with procedural room and hallway creation, HUD rendering, multiple difficulty settings, and save/load support.
- Added clickable path navigation, visibility mechanics, and monster/coin gameplay systems to support an interactive exploration experience.
- Outcome: shipped a playable Java dungeon game with deterministic world generation, interactive exploration, and persistent save/load gameplay.

Flying Drone Tracking / Autonomous Drone Following | ROS2, Python, OpenCV | Spring 2026 | GitHub | Website
- Built the ROS2 and Gazebo simulation environment for a leader-follower drone system, integrating perception, planning, and control modules end to end.
- Wrote the ArUco marker detection pipeline, tuned tracking and control parameters, and implemented low-level follower control for velocity matching and safe-distance behavior.
- Outcome: delivered a simulation testbed for autonomous drone following with target reacquisition under temporary occlusion before hardware deployment.

Languages

English: sufficient for academic and work settings | Chinese: native speaker